TOPICS IN MECHATRONICS II

3D VISION & IMAGE RECONSTRUCTION

Stereo Vision

Image Processing

ME6402

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Depth Sensors

3D Point Cloud

3D Reconstruction → Feature Extraction → Object Recognition

Project Briefing

Learning Outcomes of ME6402 (RECAP)

By the end of this course, students will be able to:

- Understand **2D and 3D vision systems**
- Understand the **geometric concepts** e.g. depth estimation and epipolar geometry
- Explain key **feature detection and matching algorithms**
- Describe **3D reconstruction techniques**
- Understand **image recognition techniques**
- Identify real-world applications of **integrated 3D vision and image recognition**

Module Schedule (RECAP)

Date	Scope
13 Jan	Module Introduction
20 Jan	3D Vision
27 Jan	Feature Detection and Matching
3 Feb	Structure from Motion (SfM) and SLAM
10 Feb	Project Briefing and Proposal Drafting **(Zoom)**
17 Feb	**Chinese New Year**
24 Feb	**Recess week**

Module Schedule (RECAP)

Date	Scope
3 Mar	**1st Presentation (Zoom)**
10 Mar	Image Recognition Techniques
17 Mar	Deep Learning for Image Recognition
24 Mar	**Project Implementation (Zoom)**
31 Mar	**Project Implementation (Zoom)**
7 Apr	**2nd Presentation (Zoom)**
14 Apr	**Project Submissions**

Aims of Applied Research Project

Through this project, students will:

- Collect and curate a set of 2D images from real-world scenes or datasets for use in 3D reconstruction
- Implement 3D reconstruction techniques to transform collected 2D images into accurate 3D models or scenes using suitable algorithms and tools
- Develop image recognition models to identify and classify objects within the reconstructed 3D images, ensuring that each group member is responsible for recognizing at least one unique object
- Integrate 3D vision and image recognition workflows to produce a seamless pipeline from data collection to 3D reconstruction and object recognition

Assessment – Applied Research Project (100%)

- Group work in teams of 4
- Group component – 65% and Individual component – 35%

Assessment Date	Assessment Component	Group	Individual	Total
3 March	Proposal Presentation (20min)			
	Presentation	5%	5%	10%
	Problem Statement and Objectives	10%	-	10%
	Literature Survey	10%	-	10%
	Proposed approach and methods	5%	5%	10%
	TOTAL	**30%**	**10%**	**40%**

Assessment – Applied Research Project (100%)

Assessment Date	Assessment Component	Group	Individual	Total
7 April	Project Presentation (30min)			
	Presentation	5%	5%	10%
	Implementation, Testing, Evaluation (3D reconstruction)	15%	-	15%
	Implementation, Testing, Evaluation (Image recognition)	-	15%	15%
	Implementation, Testing, Evaluation (Overall system)	10%	-	10%
	Reflections and Conclusions	5%	5%	10%
	TOTAL	**35%**	**25%**	**60%**

Proposal Presentation (40%)

Presentation (Group = 5% Individual = 5%)

- Communicate their proposal clearly through a presentation
- Assessed on clear explanation of the problem, literature survey and proposed methodology, involving the use of visuals where appropriate

Proposal Presentation (40%)

Problem statement and objectives (Group = 10%)

- Students will be assessed on the definition of the problem that they are solving and the articulation of the project goals
  - Clear problem statement
  - Real-world relevance of the problem
  - Well-defined objectives

Proposal Presentation (40%)

Literature survey of methods (Group = 10%)

- Research existing vision and image recognition techniques
  - 3D reconstruction techniques (e.g., SfM, SLAM)
  - Key methods in feature detection (e.g., SIFT, SURF)
  - Classification algorithms (e.g., KNN, SVM, CNN)
  - Comparison of methods (pros, cons, applications)
- Assessment on depth of research and ability to summarize key techniques with proper citations

Proposal Presentation (40%)

Proposed approach and methods (Group = 5% Individual = 5%)

- Explicitly identify selected methods from literature survey
- Assessment on the choice of methods and the rationale for selecting them to provide a logical workflow and integration of methods

Suggested Questions to be addressed (Proposal)

Problem statement and objectives

- How does this project relate to real-world applications?
- What practical or technical gap does this project aim to fill?
- What is the scope of the project?

Suggested Questions to be addressed (Proposal)

Literature Survey

- What are the common methods for 3D reconstruction from 2D images? What are the strengths and limitations of these techniques?

- What are the different approaches for object detection and recognition? What datasets are commonly used for training image recognition models?

- How are challenges such as occlusion and viewpoint variation handled?

Suggested Questions to be addressed (Proposal)

Proposed approach and methods

- What are the key tasks involved?
- What is the source of data?
- Are there any constraints or limitations (e.g. dataset size, computational resources, software tools)?

Project Presentation (60%)

Presentation (Group = 5% Individual = 5%)

- Communicate their project clearly through a presentation
- Present with a logical flow
- Concise and Focused
- Coherence with smooth transition

Project Presentation (60%)

Implementation, Testing, Evaluation (3D reconstruction) – (Group = 15%)

- Evaluates the effectiveness, quality and readability of the project's code and whether there is effective integration and use of libraries and tools

- Evaluates the robustness and reliability of the system through testing under different scenarios

- Evaluation of its performance using appropriate metrics

Project Presentation (60%)

Implementation, Testing, Evaluation (Image recognition) – (Individual = 15%)

- Evaluates the effectiveness, quality and readability of the project's code and whether there is effective integration and use of libraries and tools

- Evaluates the robustness and reliability of the system through testing under different scenarios

- Evaluation of its performance using appropriate metrics

Project Presentation (60%)

Implementation, Testing, Evaluation (Overall system) – (Group = 10%)

- Verify that 3D reconstruction and image recognition components work together
- Ensure data flows seamlessly from image source to 3D reconstruction to image recognition
- Suggest how to measure the efficiency and accuracy of the integrated system

Project Presentation (60%)

Reflections and Conclusions (Group = 5% Individual = 5%)

- Identify the key lessons learned, discuss challenges faced, and describe how the challenges were resolved
- Highlight areas for improvement
- Demonstrate skills in handling end-to-end system development

Suggested Questions to be addressed (Project)

Data Collection and Preparation

- What criteria were used to select and collect the 2D images for the project?

- How was the image dataset preprocessed to ensure quality and suitability for 3D reconstruction?

Suggested Questions to be addressed (Project)

3D Reconstruction

- What 3D reconstruction technique(s) were implemented, and why were they chosen?

- What challenges were encountered during the 3D reconstruction process, and how were they addressed?

- How accurate is the final 3D model compared to the original 2D images?

Suggested Questions to be addressed (Project)

Image Recognition

- What image recognition approach was chosen for identifying objects within the 3D model, and why?

- What were the steps involved in training, testing, and validating the image recognition model?

- How well did the model perform in recognizing the assigned objects, and what metrics were used to evaluate its performance?

Suggested Questions to be addressed (Project)

Overall System

- How were the 3D reconstruction and image recognition components integrated into a single workflow?

- What were the key challenges in integrating these components, and how were they resolved? E.g. compatibility issues

Proposal Presentation

This will be via Zoom

Date	Names		Group	Time
3 March	CHEN XINYUE FENG HAOCHEN	JIN CHENXI LI KAIXIN	1	1800 – 1820 hrs
	AUNG MYINT OO HE QIHANG	ISSAC TAY TONBY-STRANDBORG	2	1830 – 1850 hrs
	HAN HAOYU LI JIAYI	LI PEICHEN LI ZHOUJIAN	3	1900 – 1920 hrs
	GAO ZHIYU WU YULUN	YANG MINGQI ZHU YI	4	1930 – 1950 hrs
	FAN KAIWEN NI SHIWEI	YANG XULE ZHANG JIACHENG	5	2000 – 2020 hrs
	ROBIN CHAN LEONARDO PESCE	WANG ZHAN YANG YUXUAN	6	2030 – 2050 hrs

Consultations

There will be 2 sessions via Zoom

Date	Names		Group	Time
24 March	HAN HAOYU LI JIAYI	LI PEICHEN LI ZHOUJIAN	3	1800 – 1830 hrs
	GAO ZHIYU WU YULUN	YANG MINGQI ZHU YI	4	1830 – 1900 hrs
	FAN KAIWEN NI SHIWEI	YANG XULE ZHANG JIACHENG	5	1900 – 1930 hrs
	ROBIN CHAN LEONARDO PESCE	WANG ZHAN YANG YUXUAN	6	1930 – 2000 hrs
	CHEN XINYUE FENG HAOCHEN	JIN CHENXI LI KAIXIN	1	2000 – 2030 hrs
	AUNG MYINT OO HE QIHANG	ISSAC TAY TONBY-STRANDBORG	2	2030 – 2100 hrs

Consultations

There will be 2 sessions via Zoom

Date	Names		Group	Time
31 March	FAN KAIWEN NI SHIWEI	YANG XULE ZHANG JIACHENG	5	1800 – 1830 hrs
	ROBIN CHAN LEONARDO PESCE	WANG ZHAN YANG YUXUAN	6	1830 – 1900 hrs
	CHEN XINYUE FENG HAOCHEN	JIN CHENXI LI KAIXIN	1	1900 – 1930 hrs
	AUNG MYINT OO HE QIHANG	ISSAC TAY TONBY-STRANDBORG	2	1930 – 2000 hrs
	HAN HAOYU LI JIAYI	LI PEICHEN LI ZHOUJIAN	3	2000 – 2030 hrs
	GAO ZHIYU WU YULUN	YANG MINGQI ZHU YI	4	2030 – 2100 hrs

Project Presentation

This will be via Zoom

Date	Names		Group	Time
7 April	ROBIN CHAN LEONARDO PESCE	WANG ZHAN YANG YUXUAN	6	1800 – 1830 hrs
	FAN KAIWEN NI SHIWEI	YANG XULE ZHANG JIACHENG	5	1830 – 1900 hrs
	GAO ZHIYU WU YULUN	YANG MINGQI ZHU YI	4	1900 – 1930 hrs
	HAN HAOYU LI JIAYI	LI PEICHEN LI ZHOUJIAN	3	1930 – 2000 hrs
	AUNG MYINT OO HE QIHANG	ISSAC TAY TONBY-STRANDBORG	2	2000 – 2030 hrs
	CHEN XINYUE FENG HAOCHEN	JIN CHENXI LI KAIXIN	1	2030 – 2100 hrs